CONFERENCE HANDBOOK

ICCM 2026 International Conference on Celestial Mechanics

April 20-24, 2026

SHENZHEN, CHINA

The Shenzhen International Center for Mathematics, officially established on February 22, 2019, is a scientific research center funded by the Shenzhen Municipal Government and located in Southern University of Science and Technology. The scientific director is Fields Medalist Efim Zelmanov.

It aims to advance scientific research in pure mathematics, applied and computational mathematics in Shenzhen, the Greater Bay Area, and beyond.

The center will focus on important research fields in both pure and applied mathematics.

It’s primarily goal is to promote the international scientific cooperation and advance research in mathematics and interdisciplinary fields of life science, information science, engineering, finance, etc.

The SICM works in accordance with international standards for major international mathematics centers. It provides first-class research environments for researchers and visitors.

Hosting annual thematic programs, conferences, and workshops.

Welcome to the Shenzhen International Center for Mathematics!

CONFERENCE GUIDE

Conference Guide

Key on-site information for check-in, connectivity, and accommodation.

Registration

April 20, 9:00-9:30 AM
SICM 240A, Taizhou Building Second Floor

Wifi

ICMPub
Password: ICM88888

Accomodation

Shenzhen Xili Navie S Hotel (Vanke Cloud City Branch), Building 1389, Liuxian Avenue, Nanshan District
深圳西丽奈威S酒店（万科云城店）

SCHEDULE AT A GLANCE

Monday, April 20

09:30 - 10:30	**Jeff Xia** Northwestern University
10:30 - 11:00	**Coffee Break**
11:00 - 12:00	**Sergey Bolotin** Steklov Math Institute, RAS
12:00 - 14:00	**Lunch Time**
14:00 - 15:00	**Mar Giralt** Paris Observatory
15:00 - 15:30	**Coffee Break**
15:30 - 16:30	**Pau Martin** Polytechnic University of Catalonia
17:30	**Dinner Time**

SCHEDULE AT A GLANCE

Tuesday, April 21

09:30 - 10:30	**Otto van Koert**	Seoul University
10:30 - 11:00	**Coffee Break**	
11:00 - 12:00	**Alexander Batkhin**	Keldysh Institute of Applied Mathematics of RAS
12:00 - 14:00	**Lunch Time**	
14:00 - 15:00	**Gabriella Pinzari**	University of Padova
15:00 - 15:30	**Coffee Break**	
15:30 - 16:30	**Piotr Zgliczyński**	Jagiellonian University
17:30	**Dinner Time**	

SCHEDULE AT A GLANCE

Wednesday, April 22

09:30 - 10:30	**Xijun Hu** Shandong University
10:30 - 11:00	**Coffee Break**
11:00 - 12:00	**Lei Liu** Shandong University
12:00 - 14:00	**Lunch Time**
14:00 - 15:00	**Free discussions / collaboration time**
15:00 - 15:30	**Coffee Break**
15:30 - 16:30	**Free discussions / collaboration time**
17:30	**No Dinner**

Thursday, April 23

09:30 - 10:30	**Mitsuru Shibayama** Kyoto University
10:30 - 11:00	**Coffee Break**
11:00 - 12:00	**Vladislav Sidorenko** Keldysh Institute of Applied Mathematics of RAS
12:00 - 14:00	**Lunch Time**
14:00 - 15:00	**José Lamas Rodriguez** Dalian University of Technology
15:00 - 15:30	**Coffee Break**
15:30 - 16:30	**Pablo Roldan** Polytechnic University of Catalonia
17:30	**Dinner Time**

SCHEDULE AT A GLANCE

Friday, April 24

09:30 - 10:30	**Lei Zhao** Dalian University of Technology
10:30 - 11:00	**Coffee Break**
11:00 - 12:00	**Zixuan Ye** Capital Normal University
12:00 - 14:00	**Lunch Time**
14:00 - 15:00	**Cengiz Aydin** Universitat Heidelberg Bestatigte
15:00 - 15:30	**Coffee Break**
15:30 - 16:30	**Kuo-chang Chen** National Tsing Hua University
17:30	**Dinner Time**

PROGRAM DETAILS

Titles & Abstracts

Talks are presented in chronological order, with each abstract starting on a fresh page.

MONDAY, APRIL 20 | 09:30 - 10:30

On a special class of degenerate central configurations

Jeff Xia

Northwestern University

ABSTRACT

We investigate the degeneracy of the central configuration formed by a regular n-gon of equal masses together with an additional mass at the center. While degeneracy of such configurations has traditionally been studied through direct spectral computations, some partial results has been obtained in literature, a structural understanding of the origin and multiplicity of degeneracy values has remained unclear. Exploiting the dihedral symmetry D_n, we develop a representation-theoretic framework that decomposes the Hessian of $\sqrt{IU}$ into invariant blocks associated with irreducible symmetry modes. Our results provide a conceptual explanation for the multiplicity of degeneracy values and reveal that degeneracy is not an isolated phenomenon, but a structural consequence of the underlying group symmetry. This is based a joint work with Tingjie Zhou.

MONDAY, APRIL 20 11:00 - 12:00

Billiards in Celestial Mechanics

Sergey Bolotin

Steklov Math Institute, RAS

ABSTRACT

In the limit when two of the masses of the three body problem approach zero, near double collision solutions approach trajectories of a billiard type system. Nondegenerate periodic trajectories of this billiard are shadowed by near collision periodic solutions of the three body problem. Poincare named them second species solutions. Rigorous proofs of their existence were obtained much later and only for the restricted three body problem. We describe a path for constructing second species solutions for the nonrestricted problem. Then the study of the corresponding billiard is quite involved. As a model example we also discuss the two body problem on a sphere where the billiard is trivial.

MONDAY, APRIL 20 14:00 - 15:00

Chaotic phenomena to L_3 in the Restricted 3-Body Problem

Mar Giralt

Paris Observatory

ABSTRACT

The Restricted Planar Circular Three-Body Problem models the motion of a body of negligible mass under the gravitational influence of two massive bodies moving in circular orbits. It can be modeled as a two-degrees-of-freedom Hamiltonian system with five fixed points, $L_1, .., L_5$, called the Lagrange points.
We study the Lagrange point L_3 (the one point collinear with the primaries and beyond the largest one) and the invariant structures associated with it. L_3 is a saddle-center critical point, with a 1-dimensional stable and unstable manifold and a 2-dimensional center manifold folliated by a family of periodic orbits. Moreover, when the ratio between the masses of the primaries μ is small, the modulus of the hyperbolic eigenvalues are weaker, by a factor of order $\sqrt{\mu}$, than the elliptic ones.
The main result we present studies the family of periodic orbits around L_3 with Hamiltonian energy level exponentially close (with respect to $\sqrt{\mu}$) to that of L_3. In particular, we show that there exists a set of periodic orbits whose unstable and stable manifolds intersect transversally. By the Smale-Birkhoff homoclinic theorem, this implies the existence of chaotic motions (Smale's horseshoes) exponentially close to L_3 and its invariant manifolds. In addition, we also show the existence of a generic unfolding of a quadratic homoclinic tangency which leads to the existence of Newhouse domains for the RPC3BP.
To prove this result we obtain an asymptotic formula for the distance between the 1-dimensional stable and unstable manifolds of L_3. Due to the rapidly rotating dynamics, this distance is exponentially small with respect to $\sqrt{\mu}$ and, as a result, classical perturbative methods (i.e the Melnikov-Poincaré method) can not be applied.
This is a joint work with Inma Baldomá, Maciej J. Capiński and Marcel Guardia.

MONDAY, APRIL 20 15:30 - 16:30

Universality in celestial mechanics

Pau Martin

Polytechnic University of Catalonia

ABSTRACT

In this work we prove that, given any analytic area preserving map of the plane, there is a problem of celestial mechanics that approximates the map with an error arbitrarily small. More concretely, we prove that, given any compact subset K of the analytic area preserving maps of the plane and any $\varepsilon > 0$, there exists a restricted planar N-body problem such its Poincare map associated to a suitable section approximates any map in K with an error smaller than ε. The restricted planar N-body problem is associated to a suitable central configuration of N bodies, where the masses of the bodies play the role of parameters.

TUESDAY, APRIL 21 09:30 - 10:30

Symplectic dynamics and celestial mechanics.

Otto van Koert

Seoul University

ABSTRACT

In this talk, we give an overview of several techniques in symplectic dynamics. We give an outline of Floer homology, and describe its usage to understand both the existence of periodic Hamiltonian orbits, their bifurcation behavior, and persistence properties. To illustrate the usage of this theory, we apply these methods to some problems in celestial mechanics. We also describe holomorphic curve theory, a related theory, which can also be used to both understand periodic orbits and some aspects of global dynamics.

TUESDAY, APRIL 21 11:00 - 12:00

Studying Families of Asymmetric Periodic Solutions in the Hill Three-Body Problem

Alexander Batkhin

Keldysh Institute of Applied Mathematics of RAS

ABSTRACT

Families of periodic solutions in non-integrable Hamiltonian systems provide an organizing structure for invariant manifolds and, in Poincare terms, form the skeleton of the regular part of phase space. When the equations of motion admit a nontrivial discrete group of symplectomorphisms, this skeleton is typically built upon families with the highest degree of symmetry. In the classical Hill three-body problem (Hill3BP), these are the families of direct g and retrograde f satellite simple doubly symmetric orbits. Previously, the author proposed a description of second-species periodic-orbit families based on the generating solution (GS) technique. In this approach, each GS is represented by a sequence of two types of arc-solutions patched together near the singularity by a prescribed hyperbolic conic. A GS makes it possible to predict several properties of the corresponding periodic-orbit family, including its symmetry, multiplicity, asymptotic behavior of initial conditions, period, and stability index. Statistical analysis of GS classified by symmetry type shows that, beginning with sequences consisting of four or more arc-solutions, asymmetric solutions become predominant. At the same time, most periodic solutions of the Hill3BP known to date have been symmetric. This motivated a systematic search for asymmetric periodic solutions and their families. Two approaches were developed: (1) continuation from critical solutions belonging to families of single-symmetric orbits, and (2) continuation from asymmetric GS. The first approach led to two classes of asymmetric planar solutions: so-called bridges connecting families of symmetric solutions, and families continued to second-species limiting asymmetric GS. All newly found planar asymmetric families were also investigated with respect to spatial stability. As a result, several planar critical periodic solutions were identified that generate families of spatial asymmetric orbits. The second approach has not yet produced new families, apparently because GS consisting of three or more arc-solutions are extremely unstable.

TUESDAY, APRIL 21 14:00 - 15:00

On the monotonicity of the periods of the Euler problem

Gabriella Pinzari

University of Padova

ABSTRACT

We shall discuss the complete proof that periods of the Euler problem are monotone functions of the Euler integral in each component of their disconnected domain. The result had been conjectured and partly proved by H. Dullin and R. Montgomery, some year ago. The proof starts with a link with the Kepler periods and, on the analytic side, uses properties of elliptic integrals.

TUESDAY, APRIL 21

15:30 - 16:30

No infinite spin for total collisions converging to isolated central configurations in the spatial N-body problem

Piotr Zgliczyński

Jagiellonian University

ABSTRACT

In the n-body problem, when bodies tend to a total collision, then its normalized shape curve converges to the set of normalized central configurations, which has $SO(3)$ symmetry in the planar case. This leaves a possibility that the normalized shape curve tends to the set obtained by rotations of some central configuration instead of a particular point on it. This is the *infinite spin problem* which concerns the rotational behavior of total collision orbits in the n-body problem. We show that the infinite spin is not possible if the limiting shape is isolated from other connected components of the set of normalized central configurations. Our approach extends the method from recent work for total collision for the planar case by Moeckel and Montgomery. This a joint work with Gabriella Pinzari.

WEDNESDAY, APRIL 22 09:30 - 10:30

Index Theory for Singular orbits in N-body Problem

Xijun Hu

Shandong University

ABSTRACT

In the Newtonian n-body problem for solutions with arbitrary energy, which start and end either at a total collision or a parabolic/hyperbolic infinity, we prove some basic results about their Morse and Maslov indices. We compute the precise indices for some special orbits and present some applications.

WEDNESDAY, APRIL 22 11:00 - 12:00

ECH constraints and twist dynamics in the spatial isosceles three-body problem

Lei Liu

Shandong University

ABSTRACT

We study dynamical constraints arising from Embedded Contact Homology (ECH) in the spatial isosceles three-body problem. For energies below the critical level, the dynamics on the energy surface is identified with a Reeb flow on the tight three-sphere. We first obtain quantitative estimates for the Euler orbit, including monotonicity of its transverse rotation number and a strict inequality comparing its action with the contact volume. Combined with the ECH constraints of Reeb flows on the tight three-sphere with two simple periodic orbits, these estimates rule out the two-orbit scenario, thus forcing every compact energy surface below the critical level to have infinitely many periodic orbits. As a second step, this result admits a qualitative dynamical interpretation via the open book decomposition by disk-like global surfaces of section bounded by the Euler orbit. In this setting, the rotation number and the contact volume define a non-trivial twist interval which encodes the relative winding of periodic orbits. Refining Hutchings' mean action inequality, we prove that every rational number in the interior of the twist interval is realized as the mean relative winding number of two distinct periodic points of the first return map. For energies above the critical level, where the energy surface is non-compact, we prove the existence of infinitely many periodic orbits and infinitely many parabolic trajectories via twist estimates near infinity.

THURSDAY, APRIL 23 09:30 - 10:30

Existence of Really Perverse Central Configurations in the Spatial N-Body Problem

Mitsuru Shibayama

Kyoto University

ABSTRACT

We construct explicit examples of really perverse central configurations in the spatial Newtonian N-body problem. A central configuration is called really perverse if it satisfies the central configuration equations for two distinct mass distributions having the same total mass. While such configurations were previously known only in the planar case for $N \geq 474$, we prove the existence of spatial really perverse central configurations for $N = 27, \ldots, 55$.

THURSDAY, APRIL 23 11:00 - 12:00

QUASI-SATELLITE MOTION OF CELESTIAL BODIES: THE INVESTIGATION OF QUALITATIVE PROPERTIES USING WISDOM'S ADIABATIC APPROXIMATION

Vladislav Sidorenko

Keldysh Institute of Applied Mathematics of RAS

ABSTRACT

In the last decades of the XXth century it became clear that the behavior of approximate integrals of motion called adiabatic invariants is important in study of resonant effects in dynamics of celestial bodies (e.g. [1-3]). A first step of the standard scheme of the adiabatic approximation in the investigations of mean motion resonances (MMR) is averaging over the fastest dynamic process, i.e. over the orbital motion of the objects in commensurability. In averaged equations of motion one should take a subsystem that describes the process of intermediate time scale - the variation of the resonant angle. This subsystem can be interpreted as a one degree of freedom Hamiltonian system depending on other variables as slowly varying parameters. Consequently, the value of the action variable for this subsystem will be an adiabatic invariant (AI). Studying then the properties of level surfaces of AI in the subspace of the slowest variables we can draw conclusions about the secular evolution of orbits of celestial bodies in MMR.

To illustrate the opportunities of this approach, we apply it to analyze the dynamical properties of so called quasi-satellites. A quasi-satellite is an asteroid that moves in the vicinity of a planet at a distance significantly less than the distance from this planet to the Sun, and at the same time always remains outside its Hill sphere. As an example, we can mention the near-Earth asteroid 469219 Kamo'oalewa, which is expected to be visited by the Chinese space probe Tiaiwen-2. Dynamically quasi-satellite mode corresponds to 1:1 MMR. We pay special attention to the transitions between quasi-satellite modes and other forms of co-orbital motions, which can occur either deterministically or (pseudo)probabilistically.

REFERENCES

1. Neishtadt A. I., Passage through a separatrix in a resonance problem with a slowly varying parameter, J. Appl. Math. Mech. USSR, 39, 594-605 (1975).
2. Henrard J., Capture into resonance: an extension of the use of adiabatic invariants, Celest. Mech. Dyn. Astron., 27, 3-22 (1982).
3. Wisdom J., A perturbative treatment of motion near the 3/1 commensurability, Icarus, 63, 272-289 (1985).

THURSDAY, APRIL 23 | 14:00 - 15:00

Oscillatory motions, parabolic orbits and collision orbits in the planar circular restricted three-body problem

José Lamas Rodriguez

Dalian University of Technology

ABSTRACT

The planar circular restricted three body problem (PCRTBP) models the motion of a massless body under the attraction of other two bodies, the primaries, which describe circular orbits around their common center of mass. In a suitable system of coordinates, this is a two degrees of freedom Hamiltonian system. The orbits of this system are either defined for all (future or past) time or eventually go to collision with one of the primaries. For orbits defined for all time, Chazy provided a classification of all possible asymptotic behaviors, usually called final motions. By considering a sufficiently small mass ratio between the primaries, we analyze the interplay between collision orbits and various final motions and construct several types of dynamics. We show that orbits corresponding to any combination of past and future final motions can be created to pass arbitrarily close to either one of the primaries. In particular, we also establish oscillatory motions accumulating to collisions. That is, oscillatory motions in both position and velocity, meaning that as time tends to infinity, the superior limit of the position and velocity is infinity while the inferior limit of the distance to one of the primaries is zero. Additionally, we construct arbitrarily large ejection-collision orbits (orbits which experience collision in both past and future times) and periodic orbits that are arbitrarily large and get arbitrarily close to either one of the primaries. Combining these results, we construct ejection-collision orbits connecting both primaries.

THURSDAY, APRIL 23 15:30 - 16:30

Efficient shadowing of Arnold diffusion pseudo-orbits

Pablo Roldan

Polytechnic University of Catalonia

ABSTRACT

In a recent paper [GDR26], we considered the spatial R3BP around the equilibrium point L_1, and constructed diffusion pseudo-orbits with optimal diffusion time. The pseudo-orbits are obtained by iterating the so-called transition map on a suitable NHIM. They are chains of finite-time heteroclinic orbit segments, so they are not true orbits of the R3BP system. In this talk, we will explain how to construct true diffusion orbits that shadow the pseudo-orbits. We will use the transition map, the Birkhoff normal form around L_1, and a numerical parallel shooting scheme. In contrast to common shadowing lemmas used in theoretical papers, we construct shadowing orbits of similar length than the pseudo-orbits. This is joint work with A. Delshams and M. Gidea. [DGR26]: Delshams, Gidea and Roldan: Semi-analytic construction of global transfers between quasi-periodic orbits in the spatial RTBP.

FRIDAY, APRIL 24 09:30 - 10:30

Integrable Kepler billiards and the Poncelet Porism

Lei Zhao

Dalian University of Technology

ABSTRACT

In this talk I will review some recent works in the latest years concerning integrable Kepler billiards in the plane and some of their very nice geometrical properties. Some of these geometrical properties allow us to draw a direct link between their dynamics and the Poncelet porism, which in turn allows us to have a a relatively easy analysis of their dynamics in terms of elliptic curves. We shall work out an example of an integrable Kepler billiard system inside an ellipse in the plane with negative energy. Based on joint work with D. Jaud (Holzkirchen near Munich).

FRIDAY, APRIL 24 11:00 - 12:00

A variational approach to periodic orbits in the $e^- Z^{2+} e^-$ Helium

Zixuan Ye

Capital Normal University

ABSTRACT

In this article, we use variational approaches to describe generalized solutions (q_1, q_2) and critical points (z_1, z_2) of the action functional $\mathscr{B}_{av}$ for the Helium atom in the $e^- Z^{2+} e^-$ configuration with mean interaction, where (q_1, q_2) and (z_1, z_2) are related by a non-local Levi-Civita regularization introduced by Barutello, Ortega and Verzini. Additionally, we give the Lagrangian and the Hamiltonian formulations of the generalized solutions (q_1, q_2) following the framework constructed by Cieliebak, Frauenfelder and Volkov. Finally, we count the number of periodic orbits $(z_1, z_2) \in \mathscr{C}_{\mathscr{B}_{av}}$ and find the 1-to-1 correspondence between them and positive rational numbers $\mathbb{Q}_+$.

FRIDAY, APRIL 24 14:00 - 15:00

Applications of the Conley-Zehnder index to periodic orbits in the Hill problem and the CR3BP

Cengiz Aydin
Universitat Heidelberg Bestatigte

ABSTRACT

One of Hill's main contributions to celestial mechanics was his discovery of a planar direct periodic orbit with a period of one synodic lunar month in a limiting case of the circular restricted three-body problem (CR3BP) when the mass ratio of the two primaries approches zero. Since then, the Hill problem has played a fundamental role in multi-body gravitational systems. In this talk I discuss two applications of the Conley-Zehnder index to periodic orbits within the framework of Hill's modification of the CR3BP: (1) I demonstrate how to express the anomalistic and draconitic lunar months in terms of Conley-Zehnder indices and Floquet multipliers associated to Hill's lunar orbit. (2) I show how to use the Conley-Zehnder index to analyze the network structure of symmetric periodic orbit families of the Hill problem. The extensive collection of families within this problem constitutes a complex network, fundamentally comprising the so-called basic families of periodic solutions, including the orbits of the satellite g, f, the libration (Lyapunov) a, c, halo and collision $\mathcal{B}_0$ families. Since the Conley-Zehnder index leads to a grading on a topological bifurcation invariant, the local Floer homology and its Euler characteristic, the computation of those indices facilitates the construction of well-organized bifurcation graphs depicting the interconnectedness among periodic orbit families. Furthermore, I present recent results related to the Distant Retrograde Orbit (DRO) family within the Earth-Moon CR3BP.

FRIDAY, APRIL 24 15:30 - 16:30

Mass relations of some planar central configurations

Kuo-chang Chen

National Tsing Hua University

ABSTRACT

Determining relations between central configurations and masses is a challenging task in celestial mechanics. In this talk we will demonstrate some nontrivial techniques and criteria for the existence and uniqueness of some central configurations with 5 or 6 bodies. In particular, we will show criteria for rhomboidal and kite configurations, and some central configurations with non-parallel masses.

CLOSING PAGE

See you in Shenzhen.

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